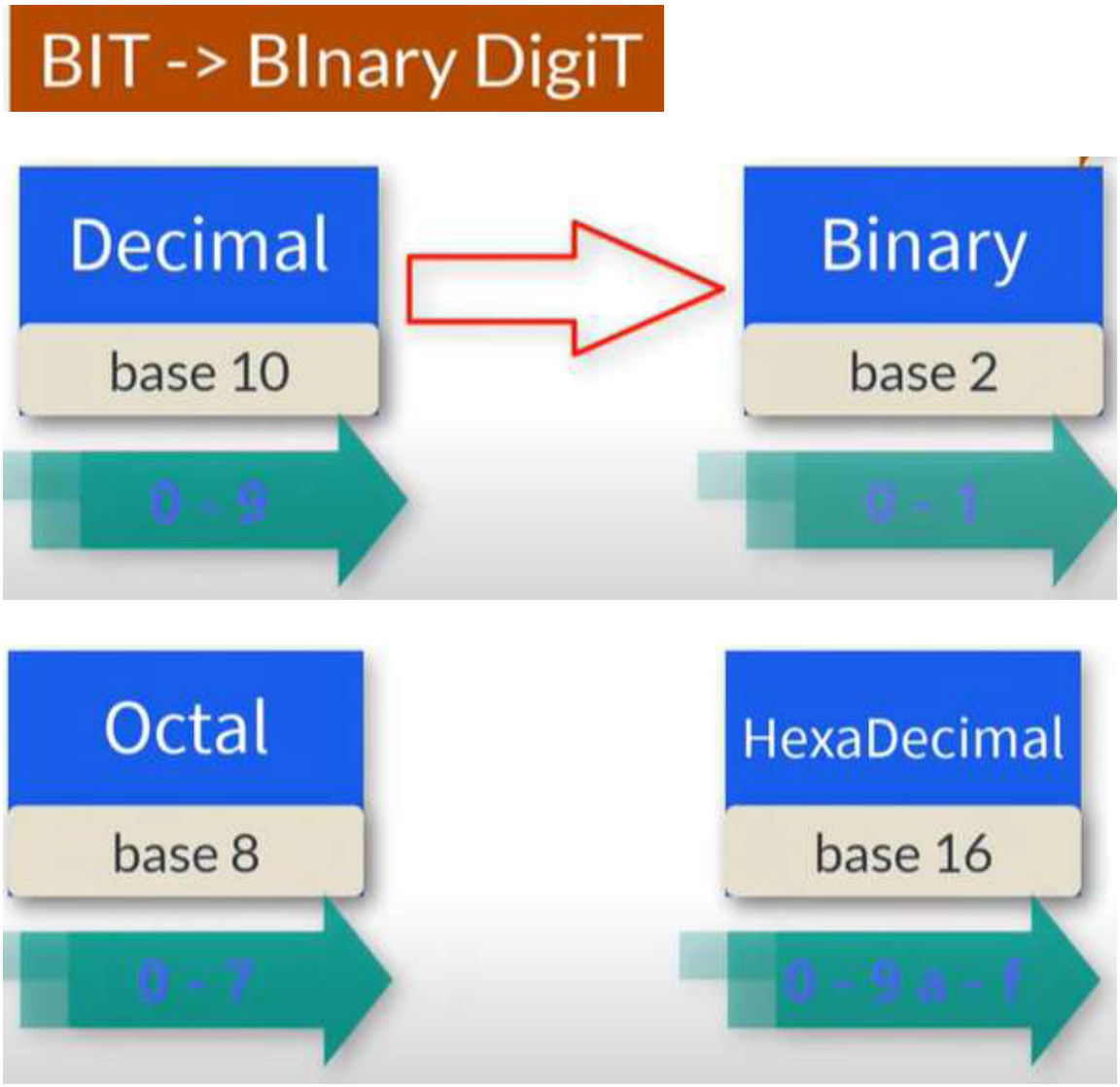
Number system

conversion of bit to binary digit




```
In [5]: 35
Out[5]: 35
In [6]: bin(35)
Out[6]: '0b100011'
In [7]: int(0b100011)
Out[7]: 35
```

In [8]:
```
bin(25)
```
Out[8]:
```
'0b11001'
```

In [9]:
```
int(0b11001)
```
Out[9]:
```
25
```

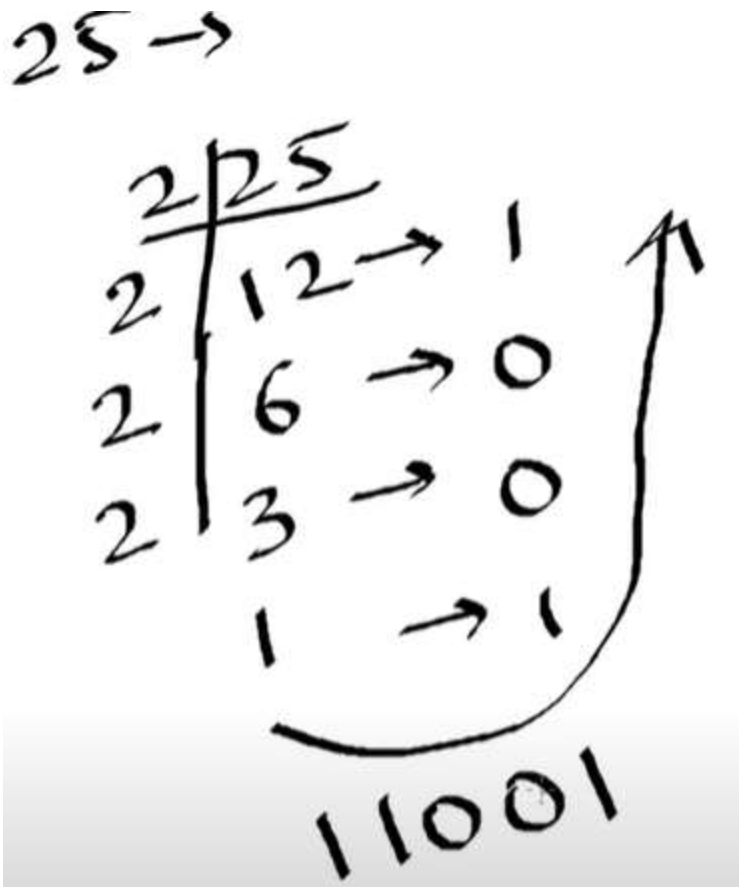
In [10]:
```
int(0b11110)
```
Out[10]:
```
30
```

In [11]:
```
bin(30)
```
Out[11]:
```
'0b11110'
```

In [13]:
```
bin(153)
```
Out[13]:
```
'0b10011001'
```

In [14]:
```
oct(25)
```
Out[14]:
```
'0o31'
```

In [15]:
```
int(0o31)
```
Out[15]:
```
25
```



In [16]:
```
0b11001
```

Out[16]: 25

In [17]:
```
int(0b11001)
```

Out[17]: 25

In [18]:
```
bin(15)
```

Out[18]: '0b1111'

In [19]:
```
int(0b1111)
```

Out[19]: 15

octal numbersystem

In [20]:
```
oct(25)
```

Out[20]: '0o31'

In [21]:
```
int(0o31)
```

```
Out[21]: 25

In [22]: oct(45)

Out[22]: '0o55'

In [23]: int(0o55)

Out[23]: 45

In [24]: oct(79)

Out[24]: '0o117'

In [25]: int(0o117)

Out[25]: 79
```

hexadecimal numbersystem

```
>>> hex(1)
'0x1'
>>> hex(2)
'0x2'
>>> hex(8)
'0x8'
>>> hex(10)
'0xa'
>>> hex(11)
'0xb'
>>> hex(256)
'0x100'
```

0x19 (Base 16)

$\Rightarrow 1 \cdot 16^0 + 9 \cdot 16$

$\Rightarrow 1 \cdot 16 + 9$

$\Rightarrow 16 + 9$

$\Rightarrow 25$

(Base 8) 0 0 3 1

$\Rightarrow 3 \cdot 8 + 1 \cdot 8^0$

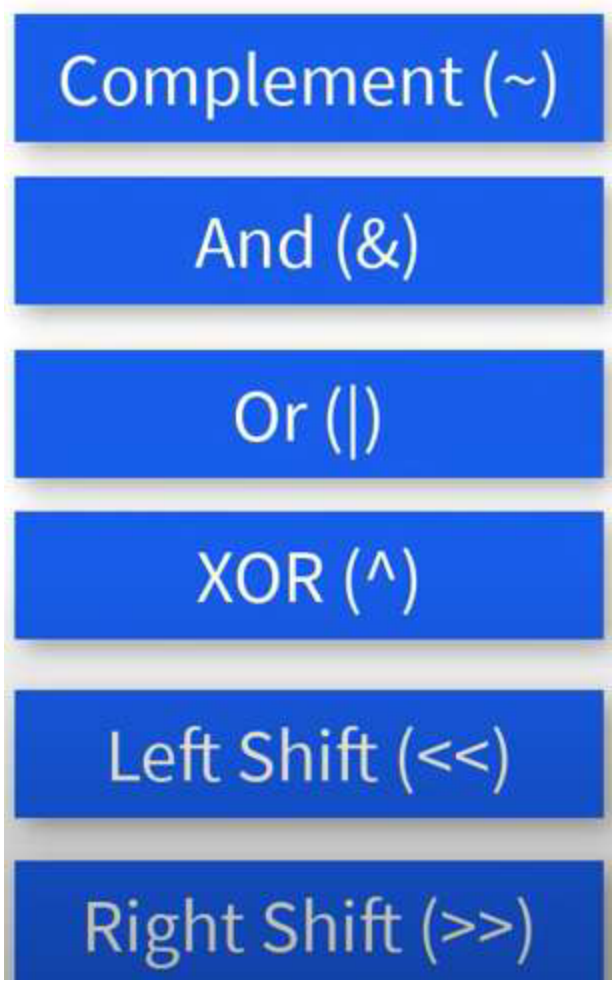
$\Rightarrow 24 + 1$

$\Rightarrow 25$

```
In [26]: hex(25)
Out[26]: '0x19'
In [27]: int(0x19)
Out[27]: 25
In [28]: int(0o31)
Out[28]: 25
In [29]: hex(19)
Out[29]: '0x13'
In [30]: int(0x13)
Out[30]: 19
In [31]: int(0x46)
Out[31]: 70
In [35]: hex(76)
Out[35]: '0x4c'
In [36]: hex(1)
Out[36]: '0x1'
```

Bitwise Operator

there are 6 operators



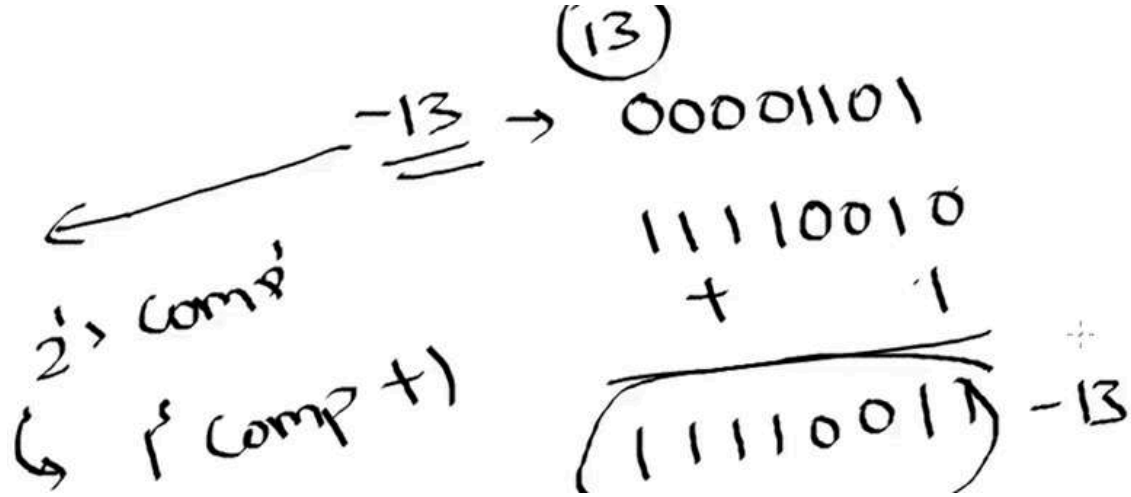
complement(~)

In [39]:
```
~12
```

Out[39]:
```
-13
```

In [40]:
```
~25
```

Out[40]:
```
-26
```




```
In [41]: ~13
Out[41]: -14
In [42]: bin(13)
Out[42]: '0b1101'
In [43]: ~46
Out[43]: -47
In [44]: ~54
Out[44]: -55
In [45]: ~10
Out[45]: -11
```

AND (&)

AND

x	y	xy
0	0	0
0	1	0
1	0	0
1	1	1

In [46]:
```
13 & 14
```

Out[46]: 12

In [47]:
```
12 & 13
```

Out[47]: 12

In [48]:
```
56 & 89
```

Out[48]: 24

OR (|)

OR

x	y	x+y
0	0	0
0	1	1
1	0	1
1	1	1

In [49]:
```
16 | 67
```

Out[49]: 83

In [50]:
```
43 | 12
```

Out[50]: 47

In [51]:
```
12 | 13
```

Out[51]: 13

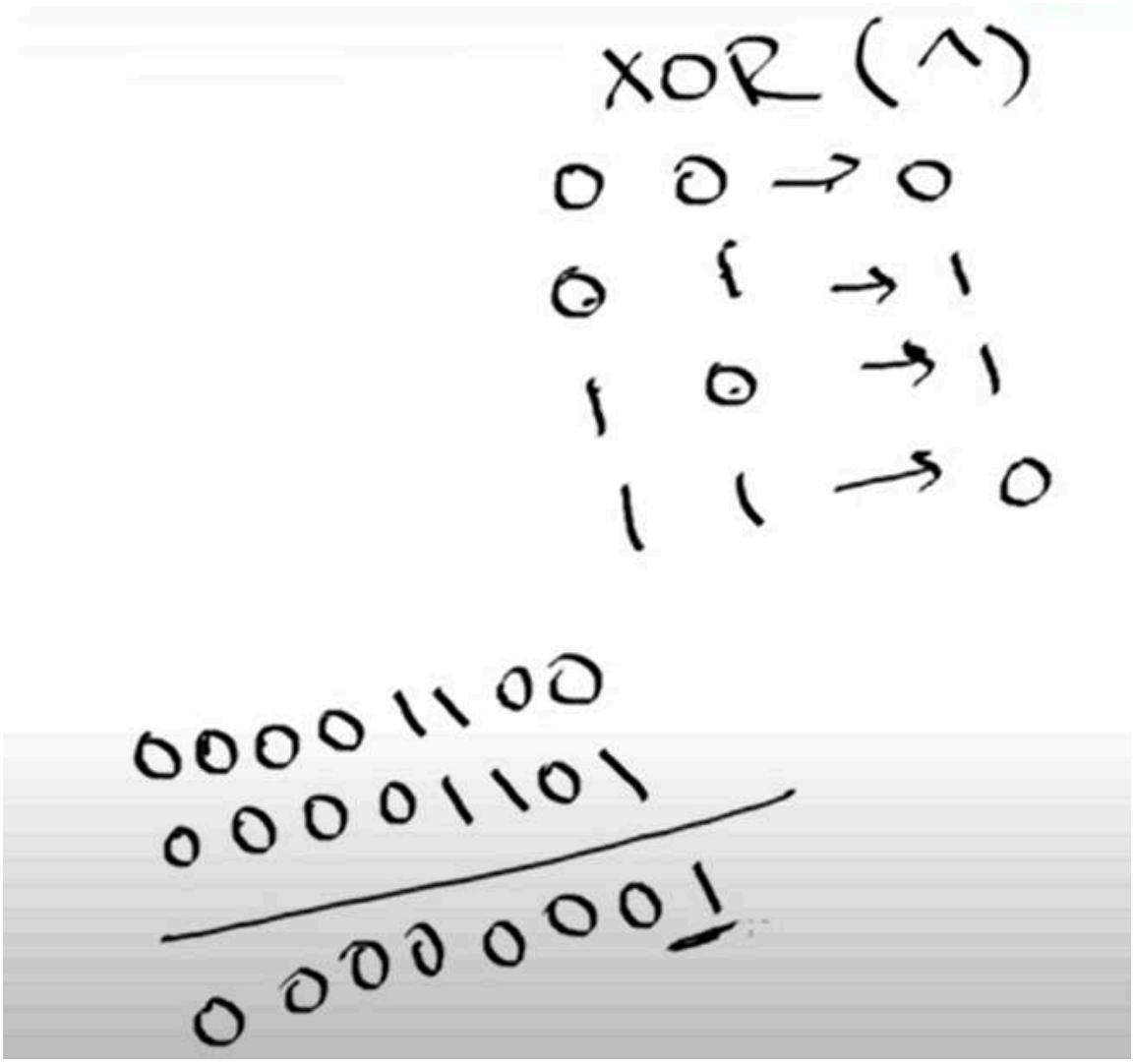
In [52]:
```
12|10
```

Out[52]: 14

In [53]:
```
100|100
```

Out[53]: 100

Exclusive or (^)



In [54]:
```
12^13
```

Out[54]: 1

In [55]:
```
25^30
```
Out[55]: 7

In [56]:
```
78^90
```
Out[56]: 20

left shift (<<)

In [57]:
```
12<<1
```
Out[57]: 24

In [58]:
```
13<<3
```
Out[58]: 104

In [59]:
```
25<<1
```
Out[59]: 50

In [60]:
```
78<<3
```
Out[60]: 624

Right shift(>>)

In [61]:
```
12>>1
```
Out[61]: 6

In [62]:
```
25>>3
```
Out[62]: 3

In [63]:
```
17>>2
```
Out[63]: 4

In [64]:
```
78>>3
```
Out[64]: 9

In [ ]: